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**Department of Computer Science.**

# SYSTEM REQUIREMENTS SPECIFICATION.

# Online Pizza Ordering System.

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# Report Overview

This report presents the System Requirements Specification (SRS) for the proposed Online Pizza Ordering System. It describes the system's purpose, main functions, quality requirements, technical requirements, business rules, and testing procedures. The report provides a clear understanding of what the system is expected to do and the conditions it must satisfy. It is organized into the following sections:

### 1. Introduction

Introduces the Online Pizza Ordering System, explains its purpose, identifies its main objectives, and describes its intended users: customers, restaurant staff, and administrators. It also defines the project scope, which focuses on ordering pizzas for restaurant pickup without delivery services.

### 2. Functional Requirements

Describes the operations the system must provide, including customer registration and login, menu browsing, pizza customization, shopping cart management, order price calculation, order tracking, order management, menu management, user and role management, sales reports, and payment processing.

### 3. Non-Functional Requirements

Defines the quality standards the system must meet, including performance, security, usability, browser compatibility, reliability, data integrity, maintainability, availability, and scalability.

### 4. Hardware Requirements

Identifies the hardware needed to access, develop, and operate the system. This section covers client-side devices, processors, memory, storage, network connectivity, server resources, and backup storage.

### 5. Software Requirements

Lists the proposed software technologies and tools for developing and deploying the system, including operating systems, front-end technologies, back-end frameworks, database management systems, development environments, web browsers, hosting services, version control tools, and testing tools.

### 6. Business Rules and Constraints

Defines the rules governing the ordering process, including required customer information, order confirmation, menu availability, price calculation, order identification, order statuses, payment verification, administrative permissions, customer privacy, and estimated preparation time.

### 7. System Verification and Testing Requirements

Explains how the system will be tested to verify that its functions operate correctly and that it satisfies the specified requirements. The tests cover authentication, pizza customization, shopping cart operations, price calculations, order tracking, user permissions, database integrity, security, payment processing where implemented, browser compatibility, and backup recovery.

# Introduction

The purpose of the Online Pizza Ordering System is to develop a web-based application that enables customers to order pizzas online. The system will allow customers to browse the menu, customize pizzas, manage shopping carts, place orders , and tell them the time the order will be ready.

The system will also provide administrative and staff functionalities for managing menu items, processing orders, monitoring order statuses, and generating sales reports.

## The system will support three main user groups:

- **Customers:** Create accounts, log in, browse the menu, customize pizzas, manage shopping carts, place orders, and track order statuses.
- **Restaurant Staff:** View incoming orders, review order details, and update order preparation statuses.
- **Administrators:** Manage menu items, prices, availability, user accounts, permissions, and sales reports.

The system supports restaurant pickup only. Delivery types re outside the scope of this project.

# Functional Requirements

Functional requirements describe the services and operations that the system must provide.

## User Registration

The system shall allow new customers to create an account by providing their name, email address, and password. The system shall validate the submitted information and ensure that the email address is not already registered. After successful registration, the customer shall be able to log in and use the features available to customer accounts.

## User Login and Logout

Each user account will have an assigned role, such as Customer, Restaurant Staff, or Administrator. When a user logs in, the system will verify their credentials and grant access only to the features permitted for their role. For example, customers can place and track their own orders, restaurant staff can manage order preparation, and administrators can manage menu items, user accounts, and sales reports. The system shall prevent users from accessing functions that are not authorized for their roles.

## Menu Browsing

The system will display the available pizzas, including their names, descriptions, images where available, sizes, crust types, toppings, prices, and availability.

## Pizza Customization

The system will allow customers to customize their pizzas by selecting from the available sizes, crust types, and additional toppings. The system also allowing display the options offered for each pizza and update its price according to the customer's selections. Only options available for the selected pizza shall be displayed.

## Shopping Cart Management

The system will allow customers to add pizzas to their shopping cart, change the quantity of each item, update their selected sizes or toppings, and remove items before confirming an order. Whenever a customer changes the cart, the system shall recalculate the subtotal and total price to reflect the updated selections.

## Order Price Calculation

The system shall calculate the total order cost based on the prices of the selected pizzas, sizes, additional toppings, and quantities. Any applicable discounts shall be included in the calculation. The system shall update the total whenever the customer changes the items or customizations in the shopping cart and display the final amount before the customer confirms the order.

## Order Tracking

The system will allow customers to view the current status of their own orders. The status may be Confirmed, Preparing, Ready for Pickup, Collected, or Cancelled. Restaurant staff shall update the order status as it moves through the preparation process.

## Order Management

The system allow authorized staff to view incoming orders and access the information necessary to prepare and process orders.

## Order Status Updates

The system allow authorized staff to update order statuses.

## Menu Management

The system allow administrators to add, edit, and remove menu items and update their descriptions, prices, sizes, toppings, and availability.

## User and Role Management

The system allow administrators to manage user accounts and assign their roles and permissions to customers and staff.

## Sales Reports

The system allow authorized administrators to view or generate reports containing order counts, sales totals, and other business information.

## Payment Processing

The system will record the payment method and payment status for each order as it in cash or online. If online payment is implemented, the system should integrate with a supported payment gateway and verify payment results securely.

# Non-Functional Requirements

Non-functional requirements define the quality standards that the system must satisfy.

## Performance

**Response Time:** The system should respond on-time to common user actions, including menu browsing, shopping cart updates, and order submission, under normal operating conditions.

**Processing Efficiency:** The system should calculate order totals and process order requests without any delays.

## Security

**Password Protection:** The system must store passwords using a password-hashing algorithm and never store passwords in plain text.

**Access Control:** The system restrict access to administrative and staff functions according to users roles.

**Customer Data Protection:** Customers can only view their own personal information and orders unless they are given permission to access others.

**Secure Communication:** The application must use HTTPS with TLS to protect information transmitted between the user's browser and the server.

**Input Validation:** The system must validate user input on the server side to reject invalid quantities, missing required fields, invalid identifiers, and unauthorized customization options.

**Database Security:** The system must use safe methods to communicate with the database to prevent SQL injection attacks. Only authorized parts of the application must be allowed to access the database.

**Web Application Security:** The system must protect against common web attacks, such as **XSS (Cross-Site Scripting)** and **CSRF (Cross-Site Request Forgery)**, and prevent unauthorized users from accessing restricted pages and features.

**Payment Security:** If online payment is implemented, the system must use a trusted payment gateway and verify payment results through a secure server-side process. The system shouldn't store full payment card numbers or card security codes.

**Session Management:** The system must keep user login sessions secure, automatically log users out after a period of inactivity, and end the session when they log out.

**Error Handling:** The system must display understandable error messages without exposing passwords, database queries, stack traces, or other sensitive technical information.

**Privacy:** The system must collect only the personal information necessary for its operations and restrict access to customer records according to user roles.

## Usability and Compatibility

**User-Friendly Interface:** The website provide a clear interface that enables customers to place pickup orders with minimal confusion.

**Responsive Design:** The website fit on desktop computers, laptops, tablets, and smartphones.

**Browser Compatibility:** The system will support modern versions of commonly used browsers, including Google Chrome, Mozilla Firefox, Microsoft Edge and Chromium-based browsers (Brave).

**Accessibility:** The system will be easy for everyone to use, with readable text, clear labels, good color contrast, and keyboard support where needed.

## Reliability and Data Integrity

**Order Accuracy:** The system maintain accurate order details, quantities and total prices throughout the ordering process.

**Duplicate Order Prevention:** The system should reduce accidental duplicate orders caused by repeated submissions or temporary network problems.

**Data Integrity:** The database should keep data accurate and ensure that users, menu items, orders, order details, and payment records are correctly connected.

**Backup and Recovery:** The system must support regular backups of essential data and provide a procedure for restoring records after data loss.

## Maintainability and Scalability

**Maintainability:** The application should use a clear structure that makes it easier to test, troubleshoot, update, and extend.

**Availability:** Availability: The system will be available during the restaurant's working hours, except during scheduled maintenance or hosting problems.

**Scalability:** The system should handle more customers and orders without becoming significantly slower.

# Hardware Requirements

The hardware requirements depend on the development environment and the expected number of users.

## Client-Side Hardware

**Devices:** Desktop computers, laptops, tablets, or smartphones.

**Processor:** A processor capable of running a modern web browser.

**Memory:** The device must have enough RAM to run the operating system and web browser smoothly.

**Storage:** The device must have enough storage space for the operating system, web browser, and system files.

**Network Connectivity:** A reliable internet connection to access the application.

## Server-Side Hardware

**Server:** A local or cloud-hosted server capable of running the web application.

**Processor:** The device must have a processor powerful enough to handle user requests smoothly.

**Memory:** The device must have enough RAM to run the system smoothly.

**Storage:** The system must have enough storage space for application files, database records, logs, and backups.

**Network Connectivity:** Reliable network access for customers and authorized staff.

**Backup Storage:** A suitable location for storing protected database backups.

# Software Requirements

The following technologies are proposed for the development and deployment of the Online Pizza Ordering System. The final selection should match the technologies chosen by the development team.

**Operating System:** Windows or Linux for the development and hosting environments.

**Front-End Technologies:** HTML5, CSS3, and JavaScript for developing the website interface. Bootstrap may be used to support responsive layouts.

**Back-End Technology:** ASP.NET Core, PHP, Node.js, or another suitable server-side framework.

**Database Management System:** Microsoft SQL Server, MySQL, or another suitable relational database management system for storing user, menu, order, and payment records.

**Development Environment:** Visual Studio or Visual Studio Code, depending on the selected technology stack.

**Web Browser:** Google Chrome, Mozilla Firefox, or Microsoft Edge for accessing and testing the system.

**Web Server and Hosting:** A compatible web server or cloud hosting service for deploying the application.

**Version Control:** Git may be used, optionally with GitHub, to manage source code and support collaboration among team members.

**External Services:** A payment gateway and notification service may be integrated if online payment and automated notifications are included in the project scope.

**Testing Tools:** Suitable tools are used for functional testing, browser compatibility testing, database testing, and security testing.

# Business Rules and Constraints

The system must follow the following business rules.

**Required Information:** Customers must provide all required information before submitting an order.

**Order Confirmation:** Customers must be able to review their selected items and total cost before confirming an order.

**Menu Availability:** Customers can order only pizzas, sizes, and toppings that are currently available.

**Price Calculation:** Order prices will be calculated according to configured menu prices, selected options, quantities, discounts, and applicable fees.

**Order Identification:** Every order will have a unique order reference like **(Order ID)**.

**Order Status:** Each order will have a valid status show its current stage of processing and pickup.

**Payment Status:** The system will confirm the payment before marking it as successful, rather than relying only on a message received in the browser **(Security Requirements)**.

**Administrative Permissions:** Only authorized administrators can modify menu prices, manage user roles, and access restricted administrative functions.

**Customer Privacy:** Customers must not be permitted to access or modify another customer's private information or orders.

**Preparation Time:** The system will determine how long it will take to prepare each order based on the restaurant's workload.

# System Verification and Testing Requirements

The system shall be tested to make sure it meets all its functional and non-functional requirements.

**Registration and Authentication Testing:** Make sure that customers can register and log in with valid information and that invalid credentials are rejected.

**Menu and Customization Testing:** Verify that available pizzas, sizes, crust types, toppings, and prices are displayed correctly.

**Shopping Cart Testing:** Verify that customers can add, modify, and remove items and that quantities and subtotals are updated correctly.

**Price Calculation Testing:** Ensure that order totals include the correct item prices, customization charges, quantities, discounts, and applicable fees.

**Order Tracking Testing:** Ensure that order statuses are displayed correctly and updated by authorized staff through preparation and pickup.

**Role and Permission Testing:** Verify that customers, staff, and administrators can access only the functions permitted by their roles.

**Database Testing:** Verify that order records and their relationships with customers and menu items remain consistent.

**Security Testing:** Test input validation, password protection, session management, access control, and relevant common web vulnerabilities in an authorized test environment.

**Payment Testing:** If online payment is implemented, verify successful, failed, and pending payment outcomes using the payment provider's test environment.

**Compatibility Testing:** Ensure that the website operates correctly on supported browsers and different screen sizes.

**Backup and Recovery Testing:** Verify that essential records can be restored from the backup.

**The system shall be tested in a suitable environment, and security tests shall only be performed on systems the development team is allowed to test.**